

Gray code

Binary counts flip many bits at once, seven to eight is a noisy edge

One-bit neighbors, convert carefully

- To encode binary to Gray, exclusive-or the value with itself shifted right by one
- Adjacent Gray codes differ by exactly one bit
- Do not do ordinary add or compare on Gray bits as if they were binary
- Async FIFOs prefer Gray pointers so only one bit is in flight across a synchronizer

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Gray code converter

Convert **binary** ↔ **Gray**, watch single-bit steps, and see why async FIFOs count in Gray.

Starter example: Width 4, binary 0000. Step the counter and compare how many bits flip in binary vs Gray — Gray always flips one.

Load starter example

Challenges 0 / 22 cleared

Quiz: to Gray: Binary→Gray formula? Answer: $B^{(B \gg 1)}$

Answer

Show hint

Check

Next

Idle

Quiz: to Gray

Quiz: property

Quiz: FIFO

Quiz: math

Starter 0

Step once

Show formula

Decode verify

Set 7

7→8 multi flip

Quiz: Gray of 1

Width 3 tour

Wrap still one

Quiz: Gray of 2

Manual set

Step back

Quiz: why Gray

Compare deltas

Encode 5

Round-trip

Quiz: name

Full insight

Core ideas

Binary → Gray

$G = B \text{ XOR } (B \gg 1)$ — each step changes one bit.

Async FIFO

Pointers cross clock domains as Gray so metastability hits one bit.

Gray code starter

learn_digital — gray code

Workbook practice

- In the workbook track, take width four
- Write binary zero through a few counts and the Gray code beside each
- Confirm each step flips one Gray bit
- Encode binary seven by hand with exclusive-or of the value and its right shift
- Note one pitfall: adding in Gray without decoding

Pitfalls to watch

- Do not treat Gray as “just another binary.” Multi-bit binary transitions are exactly what
- And remember: the browser lab is literacy
- Real CDC and FIFO designs still depend on that single-bit property across clock domains

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, build a short Gray table and one encode by hand
- When you are ready, take the short quiz, then continue to BCD